THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE FULL SYLLABUS PAPER SET 10

Full Syllabus (Chapters 1–13) | RBSE Pattern | Session 2026–27

Time: 3 Hours 15 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–28 are Very Short Answer Type (2 marks each).
4. Q. 29–36 are Short Answer Type (3 marks each).
5. Q. 37–40 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 35 to 40.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. Using atta in one trial and maida in another changes the: 1

(a) type of flour **(b)** oil temperature **(c)** time of day **(d)** size of dough

2. The first calendars were helped by watching the: 1

(a) phases of the Moon **(b)** wind direction
(c) tides of rivers **(d)** colour of leaves

3. Which part makes plant cells green? 1

(a) cell wall **(b)** nucleus **(c)** vacuole **(d)** chloroplast

4. Malaria is caused by a: 1

(a) virus **(b)** bacterium **(c)** fungus **(d)** protozoan

5. An insect carrying pathogens from one host to another is a: 1

(a) antibiotic **(b)** pathogen **(c)** vaccine **(d)** vector

6. The heat produced in a wire increases when: 1

(a) current flows for a shorter time **(b)** no current flows
(c) the wire is removed **(d)** current flows for a longer time

7. An electromagnet loses its magnetism when: 1

(a) the current is switched off **(b)** the current is switched on
(c) the iron core is added **(d)** more turns are added

8. On which listed body is the weight of 1 kg the greatest? 1

(a) Venus **(b)** Jupiter **(c)** Moon **(d)** Mars

9. A lightning conductor is a: 1

(a) rubber rod
(b) glass rod
(c) metal rod that carries charge to the ground
(d) wooden pole

10. Wind blows from: 1

(a) high pressure to low pressure **(b)** hot to cold only
(c) low pressure to high pressure **(d)** north to south always

11. The boiling point of water at atmospheric pressure is: 1

(a) 1538 °C **(b)** 0 °C **(c)** 100 °C **(d)** 133 °C

12. Silicon and boron are called: 1

(a) noble gases **(b)** alloys **(c)** alkali metals **(d)** metalloids

13. A uniform mixture is one in which: 1

(a) only solids are present **(b)** only one substance exists
(c) components are evenly distributed **(d)** components are in clear layers

14. Mass is the amount of: 1

(a) heat in an object **(b)** space occupied
(c) matter in an object **(d)** force on object

15. Which pair is correctly matched? 1

(a) Convex mirror – converging **(b)** Concave mirror – converging
(c) Plane mirror – converging **(d)** Concave lens – converging

16. The first month of the Indian National Calendar is: 1

(a) Chaitra **(b)** Asadha **(c)** Magha **(d)** Vaisakha

17. Bandipur National Park is in: 1

(a) Punjab **(b)** Gujarat **(c)** Assam **(d)** Karnataka

18. The correct sequence for energy flow is: 1

(a) herbivore → producer → carnivore **(b)** decomposer → producer → sun
(c) producer → herbivore → carnivore **(d)** carnivore → herbivore → producer

19. Which action reduces carbon emissions? 1

(a) Using public transport or cycling **(b)** Idling vehicles
(c) Burning waste **(d)** Using more plastic

20. Even small changes in oxygen level or temperature could: 1

(a) have no effect **(b)** improve life
(c) stop gravity **(d)** endanger life on Earth

SECTION — B (Very Short Answer Type) 8 × 2 = 16

Answer in about 10–15 words each.

21. How are yeasts useful in the kitchen? 2

22. How does a vaccine work? 2

23. Name two factors on which the strength of an electromagnet depends. 2

24. Why are overhead tanks placed at a height? 2

25. Define a mixture and name its components' property. 2

26. Define a solution with an example. 2

27. Name the three kinds of organisms in an ecosystem according to their food roles. 2

28. Name two greenhouse gases. 2

SECTION — C (Short Answer Type) 8 × 3 = 24

Answer in about 30–40 words each.

29. Give one example each of a yes/no observation and a measurement observation in an investigation on how fast sugar dissolves. 3

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30. Explain why pumice floats on water. 3

31. Explain how particle theory supports that smell of camphor spreads in a room. 3

32. Explain how the tilt of a mirror changes the angle of incidence for the same ray. 3

33. Why do the months in the Gregorian calendar have 30 or 31 days? 3

34. Explain the role of microorganisms in sustaining life on Earth. 3

35. Describe how Galvani's and Volta's observations led to the first battery. 3

OR

Explain why even a 'dead' battery must not be thrown in garbage. 3

36. Explain with an example how a change in one population affects others. 3

OR

What is the role of abiotic components in an ecosystem? 3

SECTION — D (Long Answer Type) — 4 × 5 = 20

Answer in about 80–100 words each.

37. Describe Alexander Fleming's discovery. How do antibiotics differ from vaccines? 5

OR

Compare acquired immunity from infection and from vaccination. 5

38. Explain local storms of India: Kalboishakhi, Bordoila and mango showers. State their usefulness. 5

OR

A tank 3 m high is on a roof. Explain why the top floor gets a weak stream. Also calculate pressure when a force of 900 N acts on 0.3 m². 5

39. Write about solid–liquid, liquid–liquid, gas–liquid and solid–solid solutions with examples. 5

OR

Explain saturated solution preparation and how to convert it into an unsaturated solution by two methods. 5

40. Describe the work of conservation: national parks, sanctuaries, biosphere reserves and community areas. 5

OR

A pond has algae, small fish, big fish and bacteria. 5
(i) Make a food chain.
(ii) Name the producer and consumers.
(iii) Role of bacteria.
(iv) Effect of removal of small fish.

END OF QUESTION PAPER — Total: 80 Marks

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